how likely, how probable, it is that we're currently in a particular branch of the wave function—which means we're assigning a probability to seeing a particular outcome of the experiment when we do look. So probability is still an essential part of quantum physics in the many-worlds interpretation; it's just that the probability isn't, strictly speaking, about outcomes of experiments, but rather about where you find yourself in the universe right now.

It's not clear whether this explanation actually works, though—it may be guilty of dualism, the idea that your mind is a nonphysical entity somehow separate from your body. Nor is it clear whether this explanation also yields the specific probabilistic predictions of quantum physics. But it is a promising idea, one among several attempts to explain this problem. Figuring out how to calculate probabilities in an infinite multiverse is one of the most pressing problems both in modern inflationary cosmology and among those who subscribe to the many-worlds interpretation. There are a variety of proposals for solving it, none of which have been widely accepted. (Some of them invoke Everett's other great mathematical love, game theory, to answer the question.) As with any open problem in science, there are no easy answers yet. But the consensus seems to be that the problem is not unequivocally solved—but that it probably can be solved, and either one of the existing promising solutions will turn out to be correct or another will be found, hopefully soon.

Despite the challenge that probability poses to the many-worlds interpretation and other multiverse theories, the most frequent objection to the idea of a multiverse (be it quantum, cosmological, or stringy) is simply its profusion of worlds. "It is hard to imagine a more radical violation of Occam's razor, the law of parsimony which urges scientists to keep entities to a minimum," complained Martin Gardner, the writer and recreational mathematician. But parsimony is in the eye of the beholder—advocates of Everett's view point out that their interpretation of quantum physics requires far fewer assumptions than any other. And arguments from simplicity, by themselves, can easily lead science astray. There are many